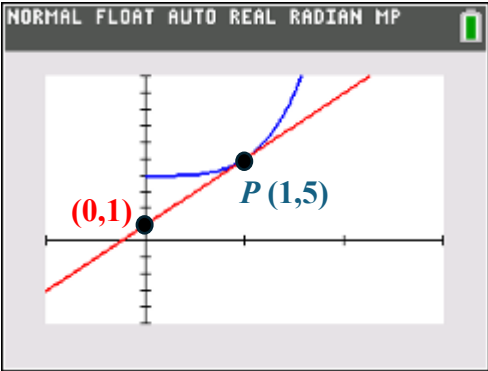
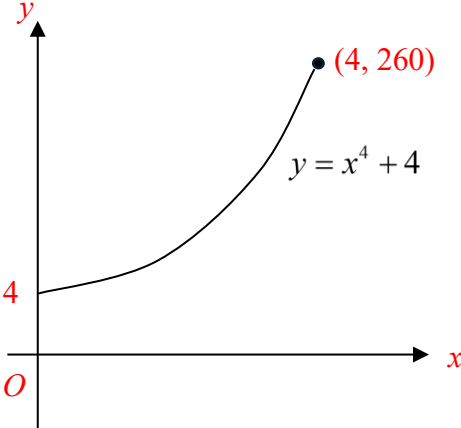


PU3 MATHEMATICS – Paper 2

Qn	Solution
1(a) [3]	Method 1 $y = f(x) \xrightarrow{x \rightarrow x-1} y = f(x-1) \xrightarrow{x \rightarrow 2x} y = f(2x-1)$ $(a, b) \qquad (a+1, b) \qquad \left(\frac{1}{2}(a+1), b\right)$ Method 2 $y = f(x) \xrightarrow{x \rightarrow 2x} y = f(2x) \xrightarrow{x \rightarrow 2\left(x-\frac{1}{2}\right)} y = f(2x-1)$ $(a, b) \qquad \left(\frac{1}{2}a, b\right) \qquad \left(\frac{1}{2}a + \frac{1}{2}, b\right)$ P becomes $P'\left(\frac{1}{2}p + \frac{1}{2}, 0\right)$, Q becomes $Q'\left(\frac{1}{2}q + \frac{1}{2}, 0\right)$ and R becomes $R'\left(\frac{1}{2}, r\right)$.
1(b) (i) [1]	The required area $= -I$ or I. <u>Transformation:</u> Reflection of the graph about the x-axis. The area in question remains the same.
1(b) (ii) [1]	The required area $= -\frac{1}{2}I$ or $\frac{1}{2} I$. <u>Transformations:</u> Scaling of the graph parallel to the y-axis by scale factor $\frac{1}{2}$. The area in question is halved. Translation of the graph in the negative x-direction by 3 units. The area in question remains the same.
1(c) [1]	$\int_0^q f'(x) \, dx = [f(x)]_0^q = f(q) - f(0)$ $= 0 - r$ $= -r$

2(i) [3]	$x = \sqrt{t}, \quad y = t^2 + 4$ $\frac{dx}{dt} = \frac{1}{2\sqrt{t}}, \quad \frac{dy}{dt} = 2t$ $\frac{dy}{dx} = \frac{2t}{\frac{1}{2\sqrt{t}}} = 4t^{\frac{3}{2}}$ Note: $t = p$ The eqn of tangent to C at $P(\sqrt{p}, p^2 + 4)$: $y - (p^2 + 4) = 4p^{\frac{3}{2}}(x - \sqrt{p})$ $y = 4p^{\frac{3}{2}}x - 3p^2 + 4$
2(ii) [2]	The eqn of tangent of C at P passes through $(0,1)$ At $(0,1)$: $1 = 4p^{\frac{3}{2}}(0) - 3p^2 + 4$ $3p^2 = 3$ $p = 1 \text{ (reject } -1 \because p \geq 0)$ When $p = 1$ $x = 1, y = 5$ $P(1,5)$ 

2 (iii) [2]	
3(a) [4]	$z + 2z^* + w = 6 + i \dots (1)$ $iz - w = 3 \Rightarrow w = iz - 3 \dots (2)$ Substitute (2) into (1): $z + 2z^* + iz - 3 = 6 + i$ $(1 + i)z + 2z^* = 9 + i$ Let $z = a + bi$. Then, $z^* = a - bi$. $(1 + i)(a + bi) + 2(a - bi) = 9 + i$ $a + bi + ai - b + 2a - 2bi = 9 + i$ $(3a - b) + (a - b)i = 9 + i$ Comparing corresponding real and imaginary parts: $3a - b = 9$ $a - b = 1$ Solving the equations simultaneously: $a = 4 \text{ and } b = 3$ $\Rightarrow z = 4 + 3i$ From (2), $w = i(4 + 3i) - 3 = -6 + 4i$ $\therefore z = 4 + 3i, w = -6 + 4i$.
3(b) [4]	<u>Method 1: Use conjugate root and factorisation</u> All the coefficients of $f(x)$ are real. Thus, $2 - i$ is also a root of $f(x) = 0$. A quadratic factor of $f(x)$ is: $[x - (2 + i)][x - (2 - i)] = [(x - 2) - i][(x - 2) + i]$ $= (x - 2)^2 - i^2$ $= x^2 - 4x + 5$

$$f(x) = x^3 + mx^2 + nx + 5 = (x + c)(x^2 - 4x + 5),$$

where c is a real constant.

Comparing constant term:

$$5c = 5 \Rightarrow c = 1$$

$$\therefore f(x) = 0 \Rightarrow x = -1 \text{ or } 2 \pm i$$

$\therefore$ The coordinates of P is $(-1, 0)$.

Method 2: Apply Factor Theorem by substituting the given root directly.

$2 + i$ is a root of $f(x) = 0$.

$$f(2 + i) = 0$$

$$(2 + i)^3 + m(2 + i)^2 + n(2 + i) + 5 = 0$$

$$2 + 11i + m(3 + 4i) + 2n + 5 + ni = 0$$

$$7 + 3m + 2n + (11 + 4m + n)i = 0$$

Comparing corresponding real and imaginary parts:

$$7 + 3m + 2n = 0 \Rightarrow 3m + 2n = -7$$

$$11 + 4m + n = 0 \Rightarrow 4m + n = -11$$

Solving the equations simultaneously:

$$m = -3 \text{ and } n = 1$$

$$\Rightarrow f(x) = x^3 - 3x^2 + x + 5$$

Using GC to solve $f(x) = 0$, $x = -1$ or $2 \pm i$.

Alternatively (instead of GC),

All the coefficients of $f(x)$ are real. Thus, $2 - i$ is also a root of $f(x) = 0$.

A quadratic factor of $f(x)$ is:

$$[x - (2 + i)][x - (2 - i)] = [(x - 2) - i][(x - 2) + i]$$

$$= (x - 2)^2 - i^2$$

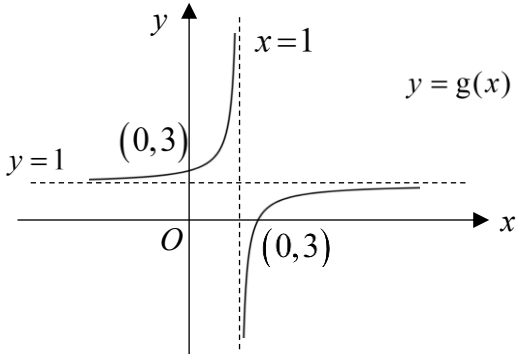
$$= x^2 - 4x + 5$$

$$\Rightarrow f(x) = x^3 - 3x^2 + x + 5 = (x + 1)(x^2 - 4x + 5).$$

$$\therefore f(x) = 0 \Rightarrow x = -1 \text{ or } 2 \pm i$$

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4(a) [1]	Given that $T_{n+2} = T_{n+1} + T_n$, and $T_1 = 3$, $T_2 = 1$ Sub $n = 1$, $T_3 = T_2 + T_1 = 1 + 3 = 4$ Sub $n = 2$, $T_4 = T_3 + T_2 = 4 + 1 = 5$																
4(b) [2]	Given $r_n = \frac{T_{n+1}}{T_n}$ $\Rightarrow r_{n+1} = \frac{T_{n+1+1}}{T_{n+1}} = \frac{T_{n+2}}{T_{n+1}}$ $r_{n+1} = \frac{T_{n+1} + T_n}{T_{n+1}}$ $r_{n+1} = 1 + \frac{T_n}{T_{n+1}}$ $r_{n+1} = 1 + \frac{1}{r_n} \text{ (shown)}$																
4(c) [3]	Since r_n is convergent to L, it implies that as $n \rightarrow \infty$, $r_n \rightarrow L$ and $r_{n+1} \rightarrow L$. Hence, $r_{n+1} = 1 + \frac{1}{r_n} \Rightarrow L = 1 + \frac{1}{L} \text{ as } n \rightarrow \infty$ $L^2 - L - 1 = 0$ $L = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2(1)}$ $L = \frac{1 \pm \sqrt{5}}{2}$ $L = \frac{1 + \sqrt{5}}{2} \text{ (rej } \frac{1 - \sqrt{5}}{2} \text{ as } r_n > 0)$																
4(d) [2]	<table border="1" style="width: 100%;"> <thead> <tr> <th style="text-align: left;">Method 1</th> <th style="text-align: left;">Method 2</th> </tr> </thead> <tbody> <tr> <td> $\sum_{k=1}^n u_k > 930$ $\frac{n}{2} [2(5) + (n-1)(2)] > 930$ $n(n+4) > 930$ From GC <table border="1" style="width: 100%;"> <tr> <td>n</td> <td>$n(n+4)$</td> </tr> <tr> <td>28</td> <td>896</td> </tr> <tr> <td>29</td> <td>957</td> </tr> </table> Least $n = 29$ </td> <td> $\sum_{k=1}^n u_k > 930$ $\sum_{k=1}^n (5 + 2(k-1)) > 930$ From GC <table border="1" style="width: 100%;"> <tr> <td>n</td> <td>$\sum_{k=1}^n (5 + 2(k-1))$</td> </tr> <tr> <td>28</td> <td>896</td> </tr> <tr> <td>29</td> <td>957</td> </tr> </table> Least $n = 29$ </td> </tr> </tbody> </table>	Method 1	Method 2	$\sum_{k=1}^n u_k > 930$ $\frac{n}{2} [2(5) + (n-1)(2)] > 930$ $n(n+4) > 930$ From GC <table border="1" style="width: 100%;"> <tr> <td>n</td> <td>$n(n+4)$</td> </tr> <tr> <td>28</td> <td>896</td> </tr> <tr> <td>29</td> <td>957</td> </tr> </table> Least $n = 29$	n	$n(n+4)$	28	896	29	957	$\sum_{k=1}^n u_k > 930$ $\sum_{k=1}^n (5 + 2(k-1)) > 930$ From GC <table border="1" style="width: 100%;"> <tr> <td>n</td> <td>$\sum_{k=1}^n (5 + 2(k-1))$</td> </tr> <tr> <td>28</td> <td>896</td> </tr> <tr> <td>29</td> <td>957</td> </tr> </table> Least $n = 29$	n	$\sum_{k=1}^n (5 + 2(k-1))$	28	896	29	957
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5(i) [3]	
5(ii) [1]	Every horizontal line $y = k$, where $k \in \mathbb{R}$, cuts the graph of $y = g(x)$ at most once. g is a one-one function. $\therefore g^{-1}$ exists.
5(iii) [3]	Let $y = g(x) = \frac{x-3}{x-1}$. $y(x-1) = x-3$ $(y-1)x = y-3$ $x = \frac{y-3}{y-1} \Rightarrow g^{-1}(y) = \frac{y-3}{y-1}$ $\therefore g^{-1}(x) = \frac{x-3}{x-1}$ From graph in part (i), $R_g = (-\infty, 1) \cup (1, \infty)$ $\therefore D_{g^{-1}} = R_g = (-\infty, 1) \cup (1, \infty)$
5(iv) [1]	For $x \in D_g$, $g(x) = \frac{x-3}{x-1}$ and $g^{-1}(x) = \frac{x-3}{x-1}$ $\therefore g(x) = g^{-1}(x)$ and hence g is self-inverse.
5(v) [2]	Since g is self-inverse, $g^2(x) = g^{-1}g(x) = gg^{-1}(x) = x$. $\therefore g^2(x) = -g^{-1}(x) \Rightarrow x = -\frac{x-3}{x-1}$ $x = -\frac{x-3}{x-1}$ $x^2 - x = -x + 3$ $x^2 = 3$ $x = \pm\sqrt{3}$

6(i)
[3]

Let X be the waiting time to get pastries from the popular bakery of a randomly chosen customer. $X \sim N(\mu, \sigma^2)$

Given $P(X < 5) = P(X > 25) = 0.10$.

Method 1

By symmetry, $\mu = \frac{5+25}{2} = 15$

$$P(X > 25) = 0.10$$

$$P\left(Z > \frac{25-15}{\sigma}\right) = 0.10$$

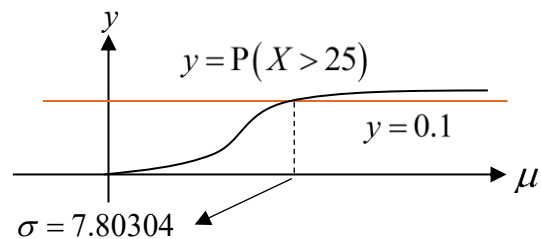
$$\frac{10}{\sigma} = 1.28155$$

$$\sigma = 7.80305$$

Alternatively,

$$P(X > 25) = 0.10$$

Using GC,



$\therefore \mu = 15, \sigma = 8$ (to the nearest minute).

Method 2

$$P(X > 25) = 0.10$$

$$P(X < 5) = 0.10$$

$$P\left(Z > \frac{25-\mu}{\sigma}\right) = 0.10$$

$$P\left(Z < \frac{5-\mu}{\sigma}\right) = 0.10$$

$$\frac{25-\mu}{\sigma} = 1.28155$$

$$\frac{5-\mu}{\sigma} = -1.28155$$

$$\mu + 1.28155\sigma = 25 \dots (1)$$

$$\mu - 1.28155\sigma = 5 \dots (2)$$

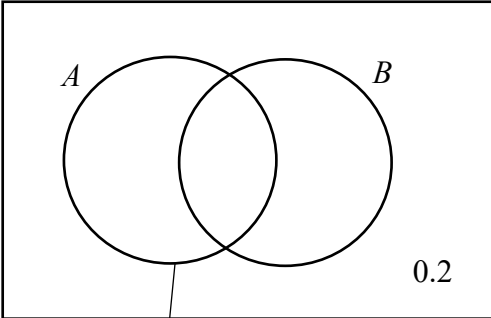
Solving (1) and (2) simultaneously,

$$\mu = 15, \sigma = 7.80305.$$

$\therefore \mu = 15, \sigma = 8$ (to the nearest minute).

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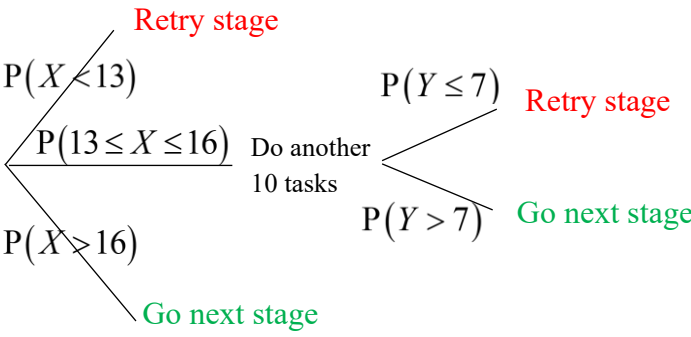
6(ii) [1]	Consider $X \sim N(15, 8^2)$. $P(X < 0) = 0.0303$. (or 0.02728 if we use $\sigma = 7.80305$) The waiting times cannot be negative but $P(X < 0)$ is about 3%, which is not insignificant. Thus, the normal distribution is not a suitable model. Note: For normal distribution: $P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.997$ i.e. 99.7%. Hence, whenever the probability of obtaining unreasonable values of x exceeds 0.003, we can conclude that the normal distribution is not suitable
7(i) [1]	The probability of obtaining a card of one colour depends solely on the colour and not any other factors. Or The probability of obtaining a card of one colour is exactly the same as the probability of obtaining another card of the same colour in the same deck.
7(ii) [4]	Method 1 $P(G = 2) = \frac{{}^2C_2 {}^{10}C_1}{{}^{12}C_3} = \frac{1}{22} \text{ (shown)}$ $P(G = 1) = \frac{{}^2C_1 {}^{10}C_2}{{}^{12}C_3} = \frac{9}{22}$ $P(G = 0) = \frac{{}^{10}C_3}{{}^{12}C_3} = \frac{12}{22} = \frac{6}{11}$ <i>Alternatively,</i> $P(G = 0) = 1 - \frac{1}{22} - \frac{9}{22} = \frac{6}{11}$ Method 2 $P(G = 2) = \left(\frac{2}{12}\right)\left(\frac{1}{11}\right)\left(\frac{10}{10}\right) \times \frac{3!}{2!} = \frac{1}{22} \text{ (shown)}$ $P(G = 1) = \left(\frac{2}{12}\right)\left(\frac{10}{11}\right)\left(\frac{9}{10}\right) \times \frac{3!}{2!} = \frac{1}{22}$ $P(G = 0) = \left(\frac{10}{12}\right)\left(\frac{9}{11}\right)\left(\frac{8}{10}\right) = \frac{6}{11}$ <i>Alternatively,</i> $P(G = 0) = 1 - \frac{1}{22} - \frac{9}{22} = \frac{6}{11}$

7(iii) [2]	<table><tr><th>G (4 pt)</th><th>S (1 pt)</th><th>Score</th><th>Probability</th></tr><tr><td>0</td><td>3</td><td>3</td><td>$\frac{12}{22} = \frac{6}{11}$</td></tr><tr><td>1</td><td>2</td><td>6</td><td>$\frac{9}{22}$</td></tr><tr><td>2</td><td>1</td><td>9</td><td>$\frac{1}{22}$</td></tr></table> <table><tr><th colspan="4">Probability Distribution of X</th></tr><tr><td>x</td><td>3</td><td>6</td><td>9</td></tr><tr><td>$P(X = x)$</td><td>$\frac{12}{22} = \frac{6}{11}$</td><td>$\frac{9}{22}$</td><td>$\frac{1}{22}$</td></tr></table>	G (4 pt)	S (1 pt)	Score	Probability	0	3	3	$\frac{12}{22} = \frac{6}{11}$	1	2	6	$\frac{9}{22}$	2	1	9	$\frac{1}{22}$	Probability Distribution of X				x	3	6	9	$P(X = x)$	$\frac{12}{22} = \frac{6}{11}$	$\frac{9}{22}$	$\frac{1}{22}$
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7(iv) [1]	$E(\text{winnings}) = 10(3) \times \frac{6}{11} + 10(6) \times \frac{9}{22} + 10(9) \times \frac{1}{22} = 45$ $\therefore$ The expected winnings is \$45.																												
	 $\frac{11}{20} = 0.55$ 0.2																												
8(i) (a) [3]	$P(B A) = \frac{P(A \cap B)}{P(A)} = \frac{4}{11}$ $P(A \cap B) = \frac{4}{11} P(A)$ $P(A \cap B) = \frac{4}{11} \left(\frac{11}{20} \right) = 0.2$ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $0.8 = 0.55 + P(B) - 0.2$ $P(B) = 0.45$																												

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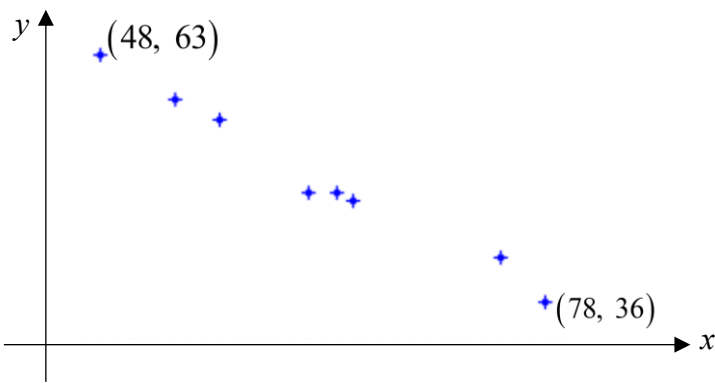
8(i) (b) [1]	<u>Method 1</u> $P(A \cap B') = P(A \cup B) - P(B) = 0.8 - 0.45 = 0.35$ <u>Method 2</u> $P(A \cap B') = P(A) - P(A \cap B) = 0.55 - 0.2 = 0.35$
8(ii) [1]	From part (i)(b), $P(A \cap B') = 0.35 \neq 0$. $\therefore A$ and B' are not mutually exclusive.
8(iii) [1]	Since B and C are independent, $P(B \cap C) = P(B)P(C) = (0.45)(0.6) = 0.27$
8(iv) [2]	From our work so far, the relevant regions are shown below. <u>Method 1</u> Let $P(A \cap B' \cap C) = x$.

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	Probability cannot be negative i.e. $0.35 - x \geq 0 \Rightarrow x \leq 0.35$ $x \geq 0$ $0.33 - x \geq 0 \Rightarrow x \leq 0.33$ $x - 0.13 \geq 0 \Rightarrow x \geq 0.13$ $\therefore 0.13 \leq P(A \cap B' \cap C) \leq 0.33$.
9(i) [2]	The two assumptions: - For each task, whether it is completed or not, is independent of that of the other tasks. - The probability that the player completes a task is the same for each task.
9(ii) [2]	$X \sim B(20, 0.7)$ The required probability = $P(13 \leq X \leq 16)$ $= P(X \leq 16) - P(X \leq 12)$ $= 0.66518$ $= 0.665$ (3 s.f.)
9(iii) (a) [3]	$Y \sim B(10, 0.8)$  P(go to the next stage) $= P(X > 16) + P(13 \leq X \leq 16)P(Y > 7)$ $= 1 - P(X \leq 16) + 0.66518[1 - P(Y \leq 7)]$ $= 0.55795$ $= 0.558$ (3 s.f.)

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9(iii) (b) [2]	The required probability $= P(X = 15 \text{don't progress to next stage})$ $= \frac{P(X = 15 \text{ and don't progress to next stage})}{P(\text{don't progress to next stage})}$ $= \frac{P(X = 15 \text{ and retry the second stage})}{P(\text{don't progress to next stage})}$ $= \frac{P(X = 15)P(Y \leq 7)}{P(\text{don't progress to next stage})}$ $= \frac{0.17886 \times 0.32220}{1 - 0.52209}$ $= 0.13037$ $= 0.130 \text{ (3 s.f.)}$
10	Let X and Y be the mass of a randomly chosen apple and pear respectively. $X \sim N(0.2, 0.05^2) \quad , \quad Y \sim N(0.3, 0.08^2)$
10(i) (a) [1]	The required probability $= P(X > 0.25)P(Y < 0.25)$ $= (0.15866)(0.26599)$ $= 0.042188$ $= 0.0422 \text{ (3 s.f.)}$
10(i) (b) [4]	The required probability = $P(X - Y \leq 0.075)$ $= P(-0.075 \leq X - Y \leq 0.075)$ $E(X - Y) = 0.2 - 0.3 = -0.1$ $\text{Var}(X - Y) = 0.05^2 + 0.08^2 = 0.0089$ $X - Y \sim N(-0.1, 0.0089)$ The required probability = $0.36371 = 0.364 \text{ (3 s.f.)}$
10(ii) [4]	The required probability $= P[5(X_1 + X_2 + \dots + X_{10}) + 7(Y_1 + Y_2 + \dots + Y_5) > 22]$ $= P[C > 22]$ $E(C) = 5[10(0.2)] + 7[5(0.3)] = 20.5$ $\text{Var}(C) = 5^2[10(0.05^2)] + 7^2[5(0.08^2)] = 2.193$ $C \sim N(20.5, 2.193)$ The required probability = $0.15555 = 0.156 \text{ (3 s.f.)}$

11(i) [1]	
11(ii) [2]	NORMAL FLOAT AUTO REAL RADIAN MP LinReg y=mx+b a=-0.8585993649 b=103.3947852 r²=0.9812291295 r=-0.9905701033 Using GC, $r = -0.991$ (3 s.f.). The value of r is close to -1. Thus, a strong negative linear correlation between the resting heart rate (x) and VO_2 max (y) is indicated.
11(iii) [3]	The regression line of y on x: $y = -0.85860x + 103.39$ $y = -0.859x + 103 \text{ (3 s.f.)}$ When $x = 60$, $y = -0.85860(60) + 103.39 = 51.874$ $\therefore$ The VO_2 max = 52 (to the nearest integer). Since $x = 60$ is within the given data range of x ($48 \leq x \leq 78$) and the value of r is close to 1. Thus, the estimate is reliable.
11(iv) [1]	When $x = 125$, $y = -3.935$. The predicted VO_2 max is negative, which is not reasonable. Moreover, the resting heart rate of 125 is not reasonable for a typical person. Thus, extrapolation is not advisable.
11(v) [3]	The value of r remains the same. $r = -0.991$ (3 s.f.) Let the adjusted resting heart rate be w. $w = x + v \Rightarrow x = w - v$ $y = -0.85860(w - v) + 103.39$ $y = -0.85860w + 0.85860v + 103.39$ $y = -0.859w + 0.859v + 103 \text{ (3 s.f.)}$ $\therefore$ The required gradient = -0.859 (3 s.f.) and the required y-intercept = $0.859v + 103$ (3 s.f.).

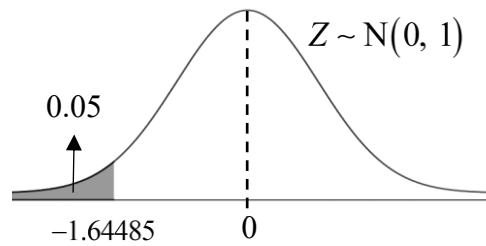
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12(i) [2]	Unbiased estimate of population mean, $\bar{x} = \frac{\sum(x-200)}{50} + 200 = \frac{2159.62}{50} + 200 = 243.1924$ Unbiased estimate of population variance, $s^2 = \frac{1}{50-1} \left[\sum(x-200)^2 - \frac{(\sum(x-200))^2}{50} \right]$ $= \frac{1}{49} \left[112081.09 - \frac{(2159.62)^2}{50} \right]$ $= 383.71$ $= 384 \text{ (3 s.f.)}$
12(ii) [5]	Let X be the reaction time, in milliseconds, of a randomly chosen student, and μ be the population mean reaction time of the students at Supernova School. $H_0 : \mu = 250$ $H_1 : \mu < 250$ Under H_0, since $n = 50$ is large, by Central Limit Theorem, $\bar{X} \sim N\left(250, \frac{383.71}{50}\right) \text{ approximately.}$ Use z-test at $\alpha = 0.01$. From GC, $p\text{-value} = 0.0069972 = 0.00700$ (3 s.f.). Since $p\text{-value} = 0.00700 < 0.01$, we reject H_0. There is sufficient evidence at 1% level of significance that the mean reaction time is reduced from 250 milliseconds.
12(iii) [1]	The sample size of 10 may not be sufficiently large for Central Limit Theorem to apply. $\bar{X}$ may not follow a normal distribution closely enough. Hence, the test in part (ii) may not be appropriate.
12(iv) [4]	Let Y be the reaction time, in milliseconds, of a randomly chosen student at Galaxy School. Method 1 $H_0 : \mu = \mu_0$ $H_1 : \mu < \mu_0$ Under H_0, $\bar{Y} \sim N\left(\mu_0, \frac{153}{40}\right)$. Test statistic, $Z = \frac{\bar{Y} - \mu_0}{\sqrt{\frac{153}{40}}} \sim N(0, 1)$. -----

Corresponding test statistic value: $z = \frac{241 - \mu_0}{\sqrt{\frac{153}{40}}}$

Critical value: -1.64485

Rejection region: $z \leq -1.64485$



Conclusion of test:

The reflex training programme was **not successful** i.e. H_0 is **not rejected** (test statistic value **does not lie in** critical region).

$$\Rightarrow \frac{241 - \mu_0}{\sqrt{\frac{153}{40}}} > -1.64485$$

$$\mu_0 < 244.22$$

$$\mu_0 < 244 \text{ (3 s.f.)}$$

The test is on finding out if the mean reaction time is reduced.

$$\therefore \mu_0 < 244 \text{ (3 s.f.)}.$$